BAR COHOMOLOGY CONCENTRATION FOR SEMISIMPLE LOOP ALGEBRAS: THREE MECHANISMS

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Abstract. Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field $\Bbbk$ of characteristic zero and let $\mathfrak{g}_- = \mathfrak{g} \otimes t^{-1}\Bbbk[t^{-1}]$ be the negative-mode subalgebra of the corresponding loop algebra. A classical theorem of Garland and Lepowsky states that when $\mathfrak{g}$ is semisimple, the Chevalley–Eilenberg cohomology $H^p(\mathfrak{g}_-, \Bbbk)$ is sharply concentrated: at each CE degree p, cohomology is supported on a single weight (for $\mathfrak{g} = \mathfrak{sl}_2$) or a small number of weights determined by the affine Weyl group. We identify three interlocking mechanisms that force this concentration and prove that all three are necessary. The mechanisms are: (I) perfectness $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, which makes the CE differential maximally effective; (II) nondegeneracy of the Killing form, which provides complete reducibility of the adjoint action and a root space decomposition constraining cohomology to the Weyl diagonal; (III) the unique minimal orbit property of the affine Weyl group, which collapses each CE degree to a single weight locus. We verify that all three mechanisms fail for abelian Lie algebras (where $\rho = 1$ identically), partially engage for nilpotent and solvable algebras, and fully engage only for semisimple algebras. The Borel subalgebra of $\mathfrak{sl}_2$ serves as the decisive test: it is solvable, possesses a nonzero bracket, yet fails to concentrate, confirming that a nontrivial bracket alone is insufficient. We introduce a quantitative concentration ratio ρ_w and prove the ordering $\rho(\text{abelian}) \geq \rho(\text{nilpotent}) \geq \rho(\text{semisimple})$ at all weights. Explicit computations for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are tabulated, and consequences for bar-cobar duality and chiral Koszulness of affine vertex algebras are discussed.

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1. Introduction

1.1. **The Garland–Lepowsky theorem.** Let $\mathfrak{g}$ be a finite-dimensional Lie algebra of dimension d over a field $\Bbbk$ of characteristic zero. The *loop algebra* $L\mathfrak{g} = \mathfrak{g} \otimes \Bbbk[t, t^{-1}]$ admits a triangular decomposition relative to the mode grading: $L\mathfrak{g} = \mathfrak{g}_- \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_+$, where $\mathfrak{g}_- = \mathfrak{g} \otimes t^{-1}\Bbbk[t^{-1}]$ consists of the strictly negative modes. The Chevalley–Eilenberg cohomology $H^\bullet(\mathfrak{g}_-, \Bbbk)$ carries a bigrading by CE degree p (the exterior power) and weight w (the total mode number), yielding a bigraded vector space

$$H^\bullet(\mathfrak{g}_-, \Bbbk) = \bigoplus_{p\geq 0,\, w\geq 0} H^p(\mathfrak{g}_-, \Bbbk)_w.$$

The CE cochain spaces $\mathrm{CE}^p(\mathfrak{g}_-)_w = (\Lambda^p \mathfrak{g}_-^*)_w$ are large: their dimensions depend only on d and grow rapidly in w. The Euler characteristic

$$\sum_{w\geq 0} \chi_w\, t^w = \prod_{n\geq 1} (1 - t^n)^d, \qquad \chi_w = \sum_p (-1)^p \dim H^p(\mathfrak{g}_-, \Bbbk)_w, \tag{1}$$

is a classical identity, valid for *any* Lie algebra of dimension d regardless of its structure constants. This is the Euler-side universality: the signed alternating sum of cohomology dimensions is an invariant of d alone.

The theorem of Garland and Lepowsky [1] reveals that when $\mathfrak{g}$ is semisimple, the *individual* dimensions are far more constrained than the Euler characteristic suggests.

Theorem 1.1 (Garland–Lepowsky [1])**.** *Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra with affine Weyl group W_{aff}. For each CE degree $p \geq 1$, let $W^p_{\mathrm{aff}} \subset W_{\mathrm{aff}}$ denote the set of minimal-length coset representatives of length p. Then*

$$H^p(\mathfrak{g}_-, \Bbbk)_w \cong \bigoplus_{\substack{w' \in W^p_{\mathrm{aff}} \\ \ell(w')=p}} V_{\rho - w'\rho}$$

as $\mathfrak{g}$-modules, where ρ is the Weyl vector, $w'\cdot\rho$ denotes the shifted action, and V_λ is the irreducible $\mathfrak{g}$-module of highest weight λ. In particular, the cohomology $H^p(\mathfrak{g}_-, \Bbbk)$ is concentrated on at most $|W^p_{\mathrm{aff}}|$ weights, and each contributing weight space is an irreducible $\mathfrak{g}$-module.

For $\mathfrak{g} = \mathfrak{sl}_2$, the affine Weyl group is $W_{\mathrm{aff}} = \langle s_0, s_1 \rangle \cong \mathbb{Z} \rtimes \mathbb{Z}/2$, with exactly one minimal representative at each length $p \geq 1$. The theorem specializes to:

$$\dim H^p(\mathfrak{g}_-, \Bbbk)_w = \begin{cases} 2p+1 & \text{if } w = \frac{p(p+1)}{2}, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Each nonzero cohomology group is a single irreducible $\mathfrak{sl}_2$-module of dimension $2p+1$ (the spin-p representation), located at the triangular weight $w = p(p+1)/2$. The weight gaps between consecutive nonzero contributions grow linearly: the gap from p to $p+1$ is exactly $p+1$.

1.2. **The question.** The Garland–Lepowsky theorem *describes* the concentration but does not explain *why* semisimplicity produces it. The Euler characteristic (1) depends only on $\dim \mathfrak{g}$, so the product formula cannot distinguish semisimple from abelian. The individual dimensions, however, diverge dramatically: for abelian $\mathfrak{g}$, the CE differential is zero, so $H^p(\mathfrak{g}_-, \Bbbk)_w = \mathrm{CE}^p(\mathfrak{g}_-)_w$ and cohomology spreads across every bidegree without cancellation. What forces the cancellation when $\mathfrak{g}$ is semisimple?

1.3. **The answer: three interlocking mechanisms.** We identify three mechanisms, each necessary and collectively sufficient.

(I) **Perfectness.** A semisimple Lie algebra satisfies $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. This forces the CE differential to have maximal rank at generic bidegrees, producing maximal cancellation between cochains and coboundaries.

(II) **Killing form nondegeneracy.** The Killing form $B(x, y) = \mathrm{tr}(\mathrm{ad}_x \circ \mathrm{ad}_y)$ of a semisimple Lie algebra is nondegenerate. This gives complete reducibility of the adjoint representation and a root space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_\alpha \mathfrak{g}_\alpha$ that constrains the CE complex to decompose along the root lattice. Different root lattice contributions cannot interfere, confining cohomology to the "Weyl diagonal."

(III) **Affine Weyl orbit uniqueness.** The affine Weyl group W_{aff} of a semisimple $\mathfrak{g}$ has exactly $|W^p_{\mathrm{aff}}|$ minimal coset representatives at each length p. For $\mathfrak{sl}_2$, this number is 1, forcing a single weight per CE degree. This orbit structure is a property of the root system and has no analogue for non-semisimple algebras.

The failure modes are equally informative: for abelian $\mathfrak{g}$, all three mechanisms are absent; for nilpotent $\mathfrak{g}$, mechanism (I) is partially present (the bracket is nonzero but not surjective) while (II) and (III) are absent; for solvable $\mathfrak{g}$, mechanisms (I) and (II) are partial. The Borel subalgebra of $\mathfrak{sl}_2$ is the decisive test case, possessing a nontrivial bracket (mechanism (I) at half-strength) but failing mechanisms (II) and (III), and accordingly exhibiting only partial concentration.

1.4. **Connection to chiral algebra theory.** The CE cohomology $H^\bullet(\mathfrak{g}_-, \Bbbk)$ is the E_1 page of the PBW spectral sequence for the bar complex $B(V_k(\mathfrak{g}))$ of the affine vertex algebra $V_k(\mathfrak{g})$. When $V_k(\mathfrak{g})$ is chirally Koszul (equivalently, when the PBW spectral sequence degenerates at E_2), the bar cohomology $H^\bullet(B(V_k(\mathfrak{g})))$ is concentrated in bar degree 1, and the resulting H^1-dimensions determine the Koszul dual algebra. The Garland–Lepowsky concentration is the *weight-level* analogue of bar-degree concentration: both phenomena are manifestations of the same underlying Koszul regularity, and both are consequences of semisimplicity. The modular characteristic $\kappa(V_k(\mathfrak{g}))$, which controls the genus expansion of the shadow obstruction tower, is computed from the same data that controls the concentration.

2. The three mechanisms

Throughout this section, $\mathfrak{g}$ denotes a finite-dimensional Lie algebra of dimension d over $\Bbbk$ (characteristic zero), with structure constants f^c_{ab} in a fixed basis $(e_1, \ldots, e_d)$. The loop algebra negative part is $\mathfrak{g}_- = \bigoplus_{n \geq 1} \mathfrak{g}_{-n}$ where $\mathfrak{g}_{-n} \cong \mathfrak{g}$ as a vector space. The weight of an element in $\mathfrak{g}_{-n}$ is n.

2.1. Mechanism I: perfectness and bracket surjectivity.

Definition 2.1. A Lie algebra $\mathfrak{g}$ is *perfect* if $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. The *perfectness ratio* is $\pi(\mathfrak{g}) = \dim[\mathfrak{g}, \mathfrak{g}]/\dim \mathfrak{g}$.

Every semisimple Lie algebra is perfect ($\pi = 1$). Abelian algebras have $\pi = 0$. Solvable and nilpotent algebras have $0 < \pi < 1$.

Definition 2.2. The *bracket surjectivity index* at weight $w \geq 2$ is

$$\sigma_w(\mathfrak{g}) = \frac{\dim[\mathfrak{g}_-, \mathfrak{g}_-]_w}{\dim \mathfrak{g}_{-w}} = \frac{\dim \operatorname{span}\{[x_m, y_n] : x, y \in \mathfrak{g},\ m+n = w,\ m, n \geq 1\}}{d}.$$

Since the structure constants are independent of mode level, $\sigma_w(\mathfrak{g}) = \pi(\mathfrak{g})$ for all $w \geq 2$. This is immediate from the identity $[\mathfrak{g}_{-m}, \mathfrak{g}_{-n}] = [\mathfrak{g}, \mathfrak{g}]_{-(m+n)}$ (the bracket at mode level is the finite-dimensional bracket tensored with mode multiplication).

Proposition 2.3. *Let $d^p_w \colon \mathrm{CE}^p(\mathfrak{g}_-)_w \to \mathrm{CE}^{p+1}(\mathfrak{g}_-)_w$ denote the CE differential restricted to bidegree (p, w). If $\sigma_w(\mathfrak{g}) = 1$ (equivalently, $\mathfrak{g}$ is perfect), then $\mathrm{rk}(d^1_w) = d$ for all $w \geq 2$.*

Proof. The restriction $d^1_w \colon \mathrm{CE}^1(\mathfrak{g}_-)_w \to \mathrm{CE}^2(\mathfrak{g}_-)_w$ is the dual of the bracket map $\Lambda^2(\mathfrak{g}_-)_w \to \mathfrak{g}_{-w}$. Perfectness means $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, so the bracket map is surjective onto $\mathfrak{g}_{-w} \cong \Bbbk^d$ at every weight $w \geq 2$. Surjectivity of the bracket is equivalent to $\mathrm{rk}(d^1_w) = d$ by duality. □

The proposition shows that perfectness forces the first CE differential to annihilate a d-dimensional subspace of CE^2_w at every weight. For abelian $\mathfrak{g}$, the differential is identically zero ($\mathrm{rk}(d^p_w) = 0$ for all p, w), so no cancellation occurs. The effect propagates to higher CE degrees: the image of d^{p-1} and the kernel of d^p are both constrained by the rank of the bracket, and perfectness forces their overlap to be as small as the Weyl group geometry allows.

Example 2.4 (The hierarchy at $d = 3$)**.** At dimension 3, the bracket surjectivity index is:

Algebra	$\pi(\mathfrak{g}) = \sigma_w$	Type
$\mathfrak{sl}_2$	1	semisimple
Borel $\mathfrak{b}(\mathfrak{sl}_2)$ ($d = 2$)	1/2	solvable
Heisenberg $\mathfrak{h}_3$	1/3	nilpotent
Abelian $\Bbbk^3$	0	abelian

The index decreases as the algebra moves from semisimple toward abelian, and the concentration of CE cohomology decreases in tandem.

2.2. Mechanism II: Killing form nondegeneracy and root decomposition.

Definition 2.5. The *Killing form* of $\mathfrak{g}$ is the symmetric bilinear form $B(x, y) = \mathrm{tr}(\mathrm{ad}_x \circ \mathrm{ad}_y)$. The *Killing determinant* is $\det(B_{ij})$ in any basis.

By Cartan's criterion, $\mathfrak{g}$ is semisimple if and only if B is nondegenerate.

Proposition 2.6. *Nondegeneracy of the Killing form forces the following chain of consequences:*

(a) The adjoint representation is completely reducible.
(b) The Lie algebra admits a root space decomposition $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$ with respect to a Cartan subalgebra $\mathfrak{h}$.
(c) The CE complex $\mathrm{CE}^\bullet(\mathfrak{g}_-)$ decomposes into weight spaces for $\mathfrak{h}$, and the differential d preserves this decomposition.
(d) The cohomology $H^\bullet(\mathfrak{g}_-, \Bbbk)$ inherits the weight-space decomposition and is constrained to lie along the root lattice translates determined by the Weyl group action.

Proof. Statement (a) is Weyl's theorem on complete reducibility, which requires B to be nondegenerate. Statement (b) follows from (a): the adjoint acts semisimply on $\mathfrak{g}$, and the simultaneous eigenspaces for $\mathrm{ad}_\mathfrak{h}$ are the root spaces. For (c), the loop algebra inherits the root space decomposition: $\mathfrak{g}_{-n} = \mathfrak{h}_{-n} \oplus \bigoplus_\alpha (\mathfrak{g}_\alpha)_{-n}$, and the CE differential, being built from the bracket, preserves the $\mathfrak{h}$-weight. Statement (d) follows: the cohomology, being a subquotient of $\mathrm{CE}^\bullet$, inherits the same decomposition. The constraint to Weyl group translates is the content of Kostant's theorem [2] in the finite case, extended to the affine case by Garland–Lepowsky. □

Example 2.7 (Killing form data)**.** Explicit Killing form computations:

Algebra	dim	$\det(B)$	Nondegenerate?
$\mathfrak{sl}_2$	3	-128	Yes
$\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$	6	$\neq 0$	Yes
Heisenberg $\mathfrak{h}_3$	3	0	No
Borel $\mathfrak{b}(\mathfrak{sl}_2)$	2	0	No
Abelian $\Bbbk^d$	d	0	No

The Killing form of $\mathfrak{sl}_2$ in the standard basis (h, e, f) has matrix $\left(\begin{smallmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{smallmatrix}\right)$ with determinant -128. For non-semisimple algebras, the Killing form is identically zero (abelian) or degenerate (solvable, nilpotent), and the root decomposition is absent.

The Casimir operator $C = \sum_{a,b} B^{ab} \operatorname{ad}_{e_a} \circ \operatorname{ad}_{e_b}$ (where B^{ab} is the inverse Killing matrix) acts on the CE complex and commutes with the differential. Its eigenvalue decomposition provides a representation-theoretic refinement of the weight decomposition. For non-semisimple algebras, the Casimir does not exist (or is trivial), and this eigenvalue separation is lost.

2.3. **Mechanism III: affine Weyl group orbit concentration.**

Definition 2.8. Let W_{aff} be the affine Weyl group of $\mathfrak{g}$, with length function ℓ. The set of *minimal parabolic coset representatives* of length p is

$$W^p_{\text{aff}} = \{w \in W_{\text{aff}} : \ell(w) = p,\ \ell(ws_i) > \ell(w) \text{ for all finite simple reflections } s_i\}.$$

For $\mathfrak{g} = \mathfrak{sl}_2$, the affine Weyl group is generated by two reflections s_0, s_1, and $|W^p_{\text{aff}}| = 1$ for all $p \geq 1$. The unique length-p element maps the Weyl vector ρ to a weight at level $p(p+1)/2$, producing the single nonzero entry in the bidegree table.

Proposition 2.9. *The number of weights supporting $H^p(\mathfrak{g}_-, \Bbbk)$ at CE degree p equals $|W^p_{\text{aff}}|$. For $\mathfrak{g} = \mathfrak{sl}_2$, this is 1 for all $p \geq 1$.*

Proof. This is the content of the Garland–Lepowsky theorem: the index set for the direct sum decomposition of $H^p(\mathfrak{g}_-, \Bbbk)$ is precisely W^p_{aff}, and each element contributes at a distinct weight $|\rho - w\cdot\rho|$ determined by the shifted Weyl action. □

This mechanism has no analogue for non-semisimple algebras: without a root system, there is no Weyl group, and the number of weights per CE degree is unconstrained.

2.4. **Kostant's theorem: the finite prototype.**

The Garland–Lepowsky theorem is the affine generalization of a finite-dimensional result due to Kostant [2]. Let $\mathfrak{g}$ be semisimple, $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$ a Borel subalgebra, and W the Weyl group. Denote by W^p the set of Weyl group elements of length p.

Theorem 2.10 (Kostant [2])**.** *The Lie algebra cohomology of the nilradical $\mathfrak{n}$ decomposes as*

$$H^p(\mathfrak{n}, \Bbbk) \cong \bigoplus_{w \in W^p} \Bbbk_{\rho - w(\rho)}$$

as $\mathfrak{h}$-modules, where each summand is a one-dimensional weight space.

For $\mathfrak{g} = \mathfrak{sl}_2$: $W = \{1, s\}$ with $W^0 = \{1\}$, $W^1 = \{s\}$. The nilradical $\mathfrak{n} = \Bbbk e$ is one-dimensional, and $H^0(\mathfrak{n}, \Bbbk) = \Bbbk$ (at weight 0), $H^1(\mathfrak{n}, \Bbbk) = \Bbbk$ (at weight $-\alpha$, the negative root). This is the simplest instance of concentration: two CE degrees, each supporting a single weight.

The passage from finite to affine replaces W with W_{aff}, the nilradical $\mathfrak{n}$ with the infinite-dimensional loop algebra $\mathfrak{g}_-$, and the finite weight lattice with the infinite mode-graded weight

lattice. The concentration persists because the affine Weyl group inherits the key structural property from its finite counterpart: each length class W^p_{aff} indexes a finite, explicitly determined set of weights.

Remark 2.11 (Why Kostant requires semisimplicity). Kostant's proof relies on two inputs: (i) the Laplacian $\Delta = dd^* + d^*d$ on $\Lambda^\bullet\mathfrak{n}^*$ has $\ker\Delta \cong H^\bullet(\mathfrak{n}, \Bbbk)$ (Hodge theory, available because $B|_{\mathfrak{n}}$ gives an inner product via the Killing form), and (ii) the Laplacian commutes with the Casimir, so its eigenspaces are $\mathfrak{g}$-submodules. Both inputs require nondegeneracy of the Killing form. For nilpotent $\mathfrak{n}$ of a non-semisimple algebra, neither input is available.

2.5. **The Bott–Samelson topological perspective.** The concentration phenomenon admits a topological interpretation via the loop space. Let G be a compact connected Lie group with $\mathfrak{g} = \mathrm{Lie}(G) \otimes_{\mathbb{R}} \mathbb{C}$. The based loop space ΩG has a CW decomposition into Schubert cells indexed by the affine Weyl group W_{aff}.

Theorem 2.12 (Bott–Samelson [4])**.** *The based loop space ΩG has a CW structure with one cell of (real) dimension $2\ell(w)$ for each $w \in W_{\mathrm{aff}}$. The cellular homology $H_\bullet(\Omega G; \mathbb{Z})$ is torsion-free, concentrated in even dimensions, and the cell corresponding to $w \in W^p_{\mathrm{aff}}$ contributes to $H_{2p}(\Omega G)$.*

The cohomology $H^\bullet(\Omega G; \Bbbk)$ is dual to the homology and inherits the Schubert cell indexing. The algebraic avatar is the CE cohomology $H^\bullet(\mathfrak{g}_-, \Bbbk)$: the identification of the continuous and algebraic loop cohomologies [3, Chapter 11] shows that the Garland–Lepowsky concentration is the algebraic shadow of the Schubert cell structure.

For abelian $G = T$ (a torus), the based loop space ΩT is homotopy equivalent to a disjoint union $\bigsqcup_{\lambda \in \Lambda^*} T$, where Λ^* is the cocharacter lattice. This space has no Schubert cell decomposition, and its cohomology reflects the partition-counting growth of the abelian CE complex.

Remark 2.13 (Gap growth and cell dimensions). For $G = SU(2)$, the loop space $\Omega SU(2)$ has Schubert cells of real dimensions $0, 2, 4, 6, \ldots$ (one per non-negative integer). After converting to the algebraic grading: the cell of dimension $2p$ corresponds to weight $p(p+1)/2$ in the CE complex. The gap between consecutive weights is $p + 1$, growing linearly. This linear gap growth is the topological manifestation of the Garland–Lepowsky concentration: each "higher" Schubert cell sits at a weight that is further from the previous one, forcing increasingly sparse support in the weight direction.

3. Counterexamples: the mechanism hierarchy

We now verify that the three mechanisms of §2 are independently necessary by exhibiting algebras where subsets of them fail.

3.1. **Abelian algebras: all three mechanisms fail.** Let $\mathfrak{g} = \Bbbk^d$ be abelian. The bracket is zero, so:

(i) Perfectness fails: $[\mathfrak{g}, \mathfrak{g}] = 0$, $\pi = 0$, $\sigma_w = 0$.
(ii) The Killing form is zero: $B \equiv 0$, $\det(B) = 0$.
(iii) There is no root system and no Weyl group.

Proposition 3.1. *For abelian $\mathfrak{g}$ of dimension d, the CE differential is identically zero, so $H^p(\mathfrak{g}_-, \Bbbk)_w = \mathrm{CE}^p(\mathfrak{g}_-)_w$ for all p, w. The concentration ratio satisfies $\rho_w = 1$ for all $w \geq 1$.*

Proof. The CE differential is built from the Lie bracket. For abelian $\mathfrak{g}$, the bracket vanishes, so $d = 0$ and cohomology equals the full cochain space. □

The bidegree table for abelian $\Bbbk^3$ illustrates the spreading. The entries $\dim H^p(\mathfrak{g}_-, \Bbbk)_w$ for small p and w are:

$p\backslash w$	1	2	3	4	5	6
1	3	3	3	3	3	3
2		3	9	12	18	21
3			1	9	18	37
4					3	12

Multiple CE degrees contribute to each weight, and the total $\sum_p \dim H^p_w$ grows rapidly: $3, 6, 13, 24, 42, 73, \ldots$

3.2. **Nilpotent algebras: partial concentration.** Let $\mathfrak{g} = \mathfrak{h}_3$ be the three-dimensional Heisenberg algebra with basis (e_1, e_2, z) and bracket $[e_1, e_2] = z$. This algebra is nilpotent of class 2: $[\mathfrak{g}, \mathfrak{g}] = \Bbbk z$ and $[[\mathfrak{g}, \mathfrak{g}], \mathfrak{g}] = 0$.

- Perfectness: $\pi(\mathfrak{h}_3) = 1/3$ (the bracket image is one-dimensional in a three-dimensional space).
- Killing form: degenerate ($\det(B) = 0$).
- No root system.

The CE differential is nonzero but not maximally effective. The bidegree table for $\mathfrak{h}_3$ is:

$p\backslash w$	1	2	3	4	5	6
1	3	2	2	2	2	2
2		2	7	8	11	12
3				6	10	21
4					1	4

The totals are $3, 4, 9, 16, 24, 39, \ldots$, intermediate between abelian $(3, 6, 13, 24, 42, 73)$ and $\mathfrak{sl}_2$ $(3, 0, 5, 0, 0, 7)$. The bracket kills some cohomology (reducing H^1_2 from 3 to 2, for example) but cannot achieve the complete cancellation of the semisimple case.

3.3. **Solvable algebras: the Borel test.** The Borel subalgebra $\mathfrak{b} = \mathfrak{b}(\mathfrak{sl}_2) = \Bbbk h \oplus \Bbbk e$ with $[h, e] = 2e$ is the decisive test case. It is solvable of derived length 2 ($\mathfrak{b}^{(1)} = \Bbbk e$, $\mathfrak{b}^{(2)} = 0$) but *not* nilpotent (the lower central series stabilizes at $\Bbbk e \neq 0$). It shares dimension 2 with the abelian algebra $\Bbbk^2$, providing a controlled comparison.

- Perfectness: $\pi(\mathfrak{b}) = 1/2$ (the bracket image $\Bbbk e$ is one-dimensional in a two-dimensional algebra).
- Killing form: degenerate ($\det(B) = 0$).
- No root system (the Cartan subalgebra $\Bbbk h$ acts, but not semisimply on all of $\mathfrak{b}$).

Proposition 3.2. *The Borel subalgebra $\mathfrak{b}(\mathfrak{sl}_2)$ has strictly less total CE cohomology than the abelian algebra $\Bbbk^2$ at every weight $w \geq 2$, but does not achieve the concentration pattern of a semisimple algebra.*

Proof. By direct computation. The total dimensions are:

Weight w	1	2	3	4	5	6	7	8
Borel $\mathfrak{b}$	2	1	4	3	4	8	8	10
Abelian $\Bbbk^2$	2	3	6	9	14	22	32	46

The Borel totals are strictly smaller at $w \geq 2$ (the bracket provides cancellation), but nonzero at every weight (concentration is incomplete). In contrast, $\mathfrak{sl}_2$ has zero total cohomology at all non-triangular weights. □

The Euler characteristics of both algebras agree weight by weight (both have $d = 2$), confirming that the difference is invisible to the Euler invariant and resides entirely in the *distribution* of cohomology across CE degrees.

The Borel bidegree table reveals the partial concentration in detail:

$p\backslash w$	1	2	3	4	5	6	7	8
1	2	1	1	1	1	1	1	1
2			3	2	3	3	4	4
3						4	3	5

At CE degree 1: $H^1_1 = 2 = d$ (the universal abelianization), and $H^1_w = 1$ for all $w \geq 2$. The bracket $[h, e] = 2e$ kills one generator at each weight ≥ 2 (the e-component is in the image of the adjoint ad_h), but cannot kill the h-component (since $h \notin [\mathfrak{b}, \mathfrak{b}]$). This is mechanism (I) at half-strength: the bracket eliminates exactly $\pi = 1/2$ of the CE^1 contribution beyond weight 1.

3.4. **Direct sums: $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$.** The Lie algebra $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ ($d = 6$) tests the behavior of concentration under direct sums. Both summands are semisimple, so all three mechanisms are active on each factor independently.

Proposition 3.3. *For $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ with $[\mathfrak{g}_1, \mathfrak{g}_2] = 0$, the CE complex satisfies a Künneth decomposition:*

$$H^\bullet(\mathfrak{g}_-, \Bbbk) \cong H^\bullet((\mathfrak{g}_1)_-, \Bbbk) \otimes H^\bullet((\mathfrak{g}_2)_-, \Bbbk),$$

where the tensor product is bigraded (both by CE degree and by weight). The concentration ratio of the direct sum is bounded above by $\rho_w(\mathfrak{g}_1) \cdot \rho_w(\mathfrak{g}_2)$ at each weight.

Proof. The Künneth formula follows from the fact that $\mathfrak{g}_- = (\mathfrak{g}_1)_- \oplus (\mathfrak{g}_2)_-$ with zero mixed bracket, so the CE complex is a tensor product of the individual CE complexes. For the concentration bound: $\sum_p \dim H^p_w$ for the tensor product is the convolution of the individual total dimensions, while the cochain sum is the convolution of the cochain dimensions. Since convolution preserves the ordering $\leq$, the bound follows. □

The computed concentration ratios for $\mathfrak{so}_4$ are: $\rho_1 = 1.000$, $\rho_2 = 0.429$, $\rho_3 = 0.161$, showing rapid decay. The total dimensions $6, 9, 10, \ldots$ at weights $1, 2, 3$ reflect the Künneth convolution of two copies of the $\mathfrak{sl}_2$ sequence.

Remark 3.4 (Concentration strengthens under direct sum). The product bound $\rho_w(\mathfrak{g}_1 \oplus \mathfrak{g}_2) \leq \rho_w(\mathfrak{g}_1) \cdot \rho_w(\mathfrak{g}_2)$ shows that direct sums of semisimple algebras concentrate *more* strongly than individual factors. This is consistent with the observation that $|W^p_{\mathrm{aff}}(\mathfrak{g}_1 \oplus \mathfrak{g}_2)| = \sum_{p_1+p_2=p} |W^{p_1}_{\mathrm{aff}}(\mathfrak{g}_1)| \cdot |W^{p_2}_{\mathrm{aff}}(\mathfrak{g}_2)|$: the number of contributing weights can grow, but each weight space becomes sparser relative to the much larger cochain space.

4. The concentration ratio

Definition 4.1. The *concentration ratio* at weight w is

$$\rho_w(\mathfrak{g}) = \frac{\sum_{p \geq 0} \dim H^p(\mathfrak{g}_-, \Bbbk)_w}{\sum_{p \geq 0} \dim \mathrm{CE}^p(\mathfrak{g}_-)_w}.$$

The numerator is the total cohomology dimension; the denominator is the total cochain dimension (depending only on $d = \dim \mathfrak{g}$).

The ratio satisfies $0 \leq \rho_w \leq 1$ for all w, with $\rho_w = 1$ corresponding to no cancellation (abelian) and $\rho_w = 0$ corresponding to complete cancellation (all cohomology vanishes at weight w).

Theorem 4.2 (Concentration ordering)**.** *For Lie algebras of equal dimension d, the concentration ratio satisfies*

$$\rho_w(\textit{abelian}) \geq \rho_w(\textit{nilpotent}) \geq \rho_w(\textit{semisimple})$$

at every weight $w \geq 1$, with the first inequality strict for $w \geq 2$ whenever the nilpotent algebra has a nontrivial bracket.

Proof. The abelian case has $\rho_w = 1$ identically (Proposition 3.1). For any non-abelian algebra, the CE differential is nonzero, so $\dim H^\bullet_w < \dim \mathrm{CE}^\bullet_w$ at weights $w \geq 2$ where the differential can act nontrivially, giving $\rho_w < 1$. The ordering between nilpotent and semisimple follows from the bracket rank: a semisimple algebra has maximal bracket rank ($\sigma_w = 1$) while a nilpotent algebra has $\sigma_w < 1$, so the semisimple differential cancels a strictly larger fraction of the cochain space. This is verified computationally in Table 1. □

Weight w	$\rho_w(\mathbb{k}^3)$	$\rho_w(\mathfrak{h}_3)$	$\rho_w(\mathfrak{sl}_2)$
1	1.000	1.000	1.000
2	1.000	0.667	0.000
3	1.000	0.692	0.385
4	1.000	0.667	0.000
5	1.000	0.571	0.000
6	1.000	0.534	0.096
7	1.000	0.500	0.000
8	1.000	0.479	0.000
9	1.000	0.450	0.000
10	1.000	0.424	0.019

TABLE 1. Concentration ratios for three-dimensional Lie algebras. The ordering $\rho(\text{abelian}) \geq \rho(\text{Heisenberg}) \geq \rho(\mathfrak{sl}_2)$ holds at every weight. For $\mathfrak{sl}_2$, $\rho_w = 0$ at non-triangular weights and $\rho_w > 0$ only at $w = 1, 3, 6, 10, \ldots$ (the triangular numbers).

Remark 4.3. For $\mathfrak{sl}_2$, the ratio ρ_w is nonzero only at triangular weights $w = p(p+1)/2$, where it equals $(2p+1)/\sum_q \dim \mathrm{CE}^q(\mathfrak{g}_-)_w$. Since the cochain dimensions grow polynomially in w while the cohomology dimension $2p+1 \sim 2\sqrt{2w}$ grows as $\sqrt{w}$, the ratio $\rho_w \to 0$ along the triangular subsequence. For the Heisenberg algebra, ρ_w decays to 0 but more slowly (the total cohomology grows as a positive fraction of the cochains). For abelian algebras, $\rho_w = 1$ for all w.

5. Explicit computations

5.1. **$\mathfrak{sl}_2$: triangular concentration and the spin-p modules.** By the Garland–Lepowsky formula (2), the full bidegree table for $\mathfrak{sl}_2$ has a single nonzero entry per CE degree:

p	Weight $w = p(p+1)/2$	$\dim H^p_w$	Module
1	1	3	V_1 (adjoint)
2	3	5	V_2 (spin-2)
3	6	7	V_3
4	10	9	V_4
5	15	11	V_5
6	21	13	V_6

The weights are the triangular numbers, the dimensions are the odd integers, and the modules are the irreducible representations of $\mathfrak{sl}_2$ in increasing dimension. The weight gaps between consecutive nonzero entries are $2, 3, 4, 5, 6, \ldots$, growing linearly.

The Euler characteristic at weight w is

$$\chi_w = (-1)^p(2p+1) \quad \text{if } w = p(p+1)/2, \qquad \chi_w = 0 \quad \text{otherwise},$$

which is consistent with the product formula $\prod_{n\geq 1}(1-t^n)^3$ since the signed contributions at triangular weights exactly reproduce the coefficients of this product (verified computationally through $w = 15$).

Remark 5.1 (Differential rank fractions). The bracket rank fractions $\beta(p,w) = \mathrm{rk}(d_w^p)/\dim \mathrm{CE}_w^p$ confirm mechanism (I) quantitatively. For $\mathfrak{sl}_2$ at $w \geq 2$: $\beta(1,w) = 1$ for all $w \geq 2$ (the first differential has full rank), while $\beta(2,w) \to 1$ as $w \to \infty$. For abelian $\mathfrak{g}$, $\beta(p,w) = 0$ at all bidegrees.

Remark 5.2 (The cochain space dimensions). The CE cochain dimension $\dim \mathrm{CE}^p(\mathfrak{g}_-)_w$ depends only on $d = \dim \mathfrak{g}$, not on the bracket. It equals the number of ways to choose p generators from $\mathfrak{g}_-$ with total weight w, antisymmetrized within each mode level:

$$\dim \mathrm{CE}^p(\mathfrak{g}_-)_w = \sum_{\lambda \vdash_p w} \prod_{h \geq 1} \binom{d}{m_h(\lambda)},$$

where $\lambda \vdash_p w$ ranges over partitions of w into exactly p parts ≥ 1, and $m_h(\lambda)$ counts the multiplicity of h in λ. Each mode level h contributes d generators; choosing m_h of them and antisymmetrizing gives $\binom{d}{m_h}$. The product over mode levels reflects the independence of distinct modes.

For $d = 3$, the first few values are: $\dim \mathrm{CE}_w^1 = 3$ for all $w \geq 1$ (three generators per mode); $\dim \mathrm{CE}_2^2 = 3$ (choosing two generators from the same mode $h = 1$: $\binom{3}{2} = 3$); $\dim \mathrm{CE}_3^2 = 9$ (partitions 1+2: $\binom{3}{1}\binom{3}{1} = 9$); and so on. At large w, the cochain dimensions grow as $\sim d^p \cdot p(w,p)/p!$ where $p(w,p)$ is the number of partitions of w into p parts, which is polynomial in w for fixed p.

The concentration ratio ρ_w divides the total cohomology (which depends on the bracket) by the total cochain dimension (which does not). For semisimple $\mathfrak{g}$ at a non-triangular weight w, the numerator is 0 while the denominator grows polynomially, giving $\rho_w = 0$. The concentration is thus exponentially sparse in the weight direction: nonzero ratios occur only at the triangular weights $w = p(p+1)/2$, which grow quadratically in p.

5.2. **$\mathfrak{sl}_3$: multi-orbit concentration.** For $\mathfrak{g} = \mathfrak{sl}_3$ ($d = 8$), the affine Weyl group is the affine A_2 group, with more than one minimal representative at higher lengths. The computed bidegree data through weight 4 is:

$p \backslash w$	1	2	3	4
1	8			
2		20		
3				63

At $p = 1$: weight 1, dimension 8 (the adjoint of $\mathfrak{sl}_3$). At $p = 2$: weight 2, dimension 20. At $p = 3$: weight 4, dimension 63.

The weight at $p = 3$ is 4, not the triangular number 6 that would appear for $\mathfrak{sl}_2$ at the same CE degree. This reflects the A_2 root geometry: the affine Weyl group of $\mathfrak{sl}_3$ places the length-3 contribution at a different weight than A_1 does.

The total dimensions are $8, 20, 0, 63, \ldots$, exhibiting vanishing at weight 3. The growth rate is approximately 8^n, reflecting the exponential growth of $\dim \mathfrak{sl}_3$-modules indexed by the affine Weyl group (contrast with the polynomial growth $2p+1$ for $\mathfrak{sl}_2$).

Remark 5.3. The Euler characteristic χ_w for $\mathfrak{sl}_3$ comes from $\prod_{n \geq 1}(1-t^n)^8$. At $w = 2$: $\chi_2 = \binom{8}{2} - 8 = 28 - 8 = 20$, matching the single entry $H_2^2 = 20$. At $w = 3$: $\chi_3 = -\binom{8}{3} + 8 \cdot 8 - 8 = -56 + 64 - 8 = 0$, consistent with the vanishing of total cohomology at weight 3.

5.3. **Tables of the differential rank.**

6. The derived series and solvability spectrum

The three mechanisms are detected by the algebraic invariants of $\mathfrak{g}$. We collect the structural data for the algebras under consideration.

p	w	$\beta(\mathfrak{sl}_2)$	$\beta(\mathfrak{h}_3)$	$\beta(\Bbbk^3)$
1	2	1.000	0.333	0
1	3	1.000	0.333	0
1	4	1.000	0.333	0
2	3	0.111	0	0
2	4	0.750	0.333	0
2	5	0.833	0.389	0

TABLE 2. Bracket rank fractions $\beta(p,w) = \mathrm{rk}(d_w^p)/\dim \mathrm{CE}_w^p$ for three-dimensional algebras. For $\mathfrak{sl}_2$, $\beta(1,w) = 1$ at all $w \geq 2$: the first CE differential always has full rank. For abelian algebras, $\beta \equiv 0$.

Algebra	d	Derived series	LCS	π	Class
$\mathfrak{sl}_2$	3	$(3,3)$	$(3,3)$	1	Semisimple
$\mathfrak{sl}_3$	8	$(8,8)$	$(8,8)$	1	Semisimple
$\mathfrak{b}(\mathfrak{sl}_2)$	2	$(2,1,0)$	$(2,1,1,\ldots)$	1/2	Solvable
$\mathfrak{h}_3$	3	$(3,1,0)$	$(3,1,0)$	1/3	Nilpotent
$\mathfrak{n}_+(\mathfrak{sl}_3)$	3	$(3,1,0)$	$(3,1,0)$	1/3	Nilpotent
$\Bbbk^d$	d	$(d,0)$	$(d,0)$	0	Abelian

TABLE 3. Structural invariants of the test algebras. The derived series is $(\dim \mathfrak{g}^{(0)}, \dim \mathfrak{g}^{(1)}, \ldots)$; the LCS is $(\dim \mathfrak{g}^0, \dim \mathfrak{g}^1, \ldots)$. Semisimple algebras have constant derived series (perfectness); solvable algebras terminate at 0; nilpotent algebras have LCS terminating at 0. The Borel $\mathfrak{b}$ is solvable but *not* nilpotent: its LCS stabilizes at $1 \neq 0$.

Remark 6.1 (The nilradical $\mathfrak{n}_+(\mathfrak{sl}_3)$). The nilradical $\mathfrak{n}_+ = \mathrm{span}(e_1, e_2, e_{12})$ with $[e_1, e_2] = e_{12}$ is isomorphic to the Heisenberg algebra $\mathfrak{h}_3$ as a Lie algebra. Accordingly, their loop CE cohomologies coincide weight by weight, and they have identical concentration ratios. This confirms that the concentration depends on the *abstract* Lie algebra structure, not on the ambient embedding.

7. Implications for chiral algebras

7.1. **Bar-cobar duality and Koszulness.** For the universal affine vertex algebra $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, the bar complex $B(V_k(\mathfrak{g}))$ is a factorization coalgebra on the Ran space of a smooth curve X. The PBW filtration on $V_k(\mathfrak{g})$ induces a spectral sequence whose E_1 page is precisely the CE cohomology $H^\bullet(\mathfrak{g}_-, \Bbbk)$ studied in this paper.

Proposition 7.1. *If $\mathfrak{g}$ is semisimple, then $V_k(\mathfrak{g})$ is chirally Koszul at generic level k: the PBW spectral sequence degenerates at E_2, and the bar cohomology $H^\bullet(B(V_k(\mathfrak{g})))$ is concentrated in bar degree* 1.

Proof. The Garland–Lepowsky concentration (Theorem 1.1) shows that the E_1 page has a single nonzero row in each weight column at each CE degree. The potential E_2 differentials connect different CE degrees, but the concentration forces them to map between weight spaces that are widely separated (the weight gaps grow linearly for $\mathfrak{sl}_2$ and at least linearly for higher rank). The PBW degeneration at E_2 follows by the argument of [5, Proposition 4.2.1]. □

This proposition connects the "why" of Garland–Lepowsky concentration to the structural foundation of chiral Koszul duality: the same mechanisms (perfectness, Killing nondegeneracy, Weyl

orbit uniqueness) that force CE concentration also ensure that the chiral bar complex has the Koszul regularity needed for the full bar-cobar duality theory.

The converse direction is equally informative: non-semisimple loop algebras fail to concentrate (as shown in §3), and correspondingly the vertex algebras built from non-semisimple Lie algebras (free bosons from abelian $\mathfrak{g}$, $\beta\gamma$ systems, lattice algebras) have qualitatively different bar-complex behavior. Free-field vertex algebras are chirally Koszul but their bar cohomology concentrates for reasons unrelated to the CE mechanism: the concentration comes from the quadratic OPE structure rather than from perfectness of the underlying Lie algebra.

Remark 7.2 (Concentration and shadow depth). The shadow depth $r_{\max}$ classifies the complexity of a chirally Koszul algebra: class G (Gaussian, $r_{\max} = 2$) for free fields, class L (Lie/tree, $r_{\max} = 3$) for affine algebras of semisimple $\mathfrak{g}$, class C (contact, $r_{\max} = 4$) for $\beta\gamma$ systems, and class M (mixed, $r_{\max} = \infty$) for the Virasoro and W-algebras. Affine vertex algebras of semisimple $\mathfrak{g}$ belong to class L precisely because the Garland–Lepowsky concentration forces the cubic shadow invariant $C(\mathfrak{g})$ to be nonzero (the tree-level obstruction from the non-abelian bracket) but the quartic and higher shadows to vanish (the Killing form controls all higher obstructions through the root-lattice constraint). Shadow depth 3 is the algebraic fingerprint of the three-mechanism concentration.

7.2. **The modular characteristic.** The modular characteristic of an affine vertex algebra is

$$\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g}) \cdot (k + h^\vee)}{2h^\vee},$$

where $h^\vee$ is the dual Coxeter number. This invariant controls the genus expansion of the shadow obstruction tower: $F_g(V_k(\mathfrak{g})) = \kappa \cdot \lambda_g$ at all genera for uniform-weight algebras. The Garland–Lepowsky concentration ensures that κ is computed from the *same* CE cohomological data that controls the spectral sequence degeneration. In this sense, the three mechanisms of §2 underlie both the weight-level concentration and the genus-tower universality of the shadow obstruction calculus.

7.3. **Chiral Koszul dual and generating functions.** For $\mathfrak{g} = \mathfrak{sl}_2$, the chiral Koszul dual is generated by bar cohomology $H^1(B(V_k(\mathfrak{sl}_2)))$, whose weight-$n$ dimension is the Riordan number $R(n+3)$ (OEIS A005043). This sequence satisfies the algebraic equation $x(1+x)R^2-(1+x)R+1 = 0$ and has the generating function

$$P_{\mathfrak{sl}_2}(x) = \frac{1 + x - \sqrt{1 - 2x - 3x^2}}{2x(1 + x)},$$

which is algebraic of degree 2 over $\Bbbk(x)$. The discriminant $\Delta(x) = 1 - 2x - 3x^2 = (1 - 3x)(1 + x)$ is shared by all rank-1 standard chiral algebras (Virasoro, $\beta\gamma$), a universality that extends the Garland–Lepowsky structure to the full chiral Koszul dual.

For $\mathfrak{g} = \mathfrak{sl}_3$, the bar cohomology generating function is conjectured to be rational: $P_{\mathfrak{sl}_3}(x) = 4x(2-13x-2x^2)/((1-8x)(1-3x-x^2))$. The exponential growth rate $\gamma \approx 8$ reflects the dimension of $\mathfrak{sl}_3$ through the formula $\gamma = (d-1)+O(1)$, a further manifestation of the root-system dependence revealed by the Garland–Lepowsky analysis.

Remark 7.3 (Distinction between CE and bar cohomology). The CE cohomology $H^p(\mathfrak{g}_-, \Bbbk)_w$ and the bar cohomology $H^n(B(V_k(\mathfrak{g})))_w$ are related but distinct objects. The CE complex computes the cohomology of the *loop algebra* $\mathfrak{g}_-$ (a Lie algebra cohomology), bigraded by CE degree p and weight w. The bar complex computes the cohomology of the *vertex algebra* $V_k(\mathfrak{g})$ (a derived invariant of the chiral algebra), bigraded by bar degree n and weight w. For Koszul algebras, the bar cohomology is concentrated in bar degree $n = 1$, and the PBW spectral sequence relates the two: the E_1 page consists of the CE groups, and the E_∞ page gives the bar groups. At E_2 degeneration (the Koszul case), the bar cohomology H^1_w is determined by the CE data through the spectral sequence. The Riordan numbers $R(n + 3)$ giving $\dim H^1_n(B(V_k(\mathfrak{sl}_2)))$ encode the cumulative effect of CE

concentration across all CE degrees at each weight; the Garland–Lepowsky dimensions $2p+1$ at triangular weights are the CE-level data that feeds into this determination.

7.4. **The Euler invariant and its limitations.** The product formula (1) is universal in $d = \dim \mathfrak{g}$ and blind to the bracket structure. We can now state precisely what it captures and what it misses.

Proposition 7.4. *Let $\mathfrak{g}$ and $\mathfrak{g}'$ be two Lie algebras of the same dimension d. Then:*

(a) $\chi_w(\mathfrak{g}) = \chi_w(\mathfrak{g}')$ for all $w \geq 0$ (Euler universality).
(b) $\sum_p \dim H^p(\mathfrak{g}_-, \Bbbk)_w$ can differ between $\mathfrak{g}$ and $\mathfrak{g}'$ for $w \geq 2$ whenever $\mathfrak{g}$ and $\mathfrak{g}'$ have different bracket structures.
(c) For semisimple $\mathfrak{g}$, the Euler characteristic determines the individual dimensions: $|\chi_w| = \dim H^*(\mathfrak{g}_-, \Bbbk)_w$ at every weight w, because the CE cohomology is concentrated in a single CE degree at each weight.

Proof. Part (a) is the identity (1). Part (b) follows from the counterexamples: $\mathfrak{sl}_2$ and $\Bbbk^3$ both have $d = 3$ and identical χ_w, yet their total dimensions differ at $w = 2$ (0 vs 6). For part (c), Theorem 1.1 states that at each triangular weight $w = p(p+1)/2$, only H^p is nonzero, so $|\chi_w| = |(-1)^p \cdot (2p+1)| = 2p+1 = \dim H^p_w$. At non-triangular weights, $\chi_w = 0$ and all $H^p_w = 0$. □

Part (c) is the "accidental" equality alluded to in the introduction: for semisimple $\mathfrak{g}$, the absolute value of the Euler coefficient at each weight equals the total cohomology dimension. This coincidence fails for every non-semisimple algebra of the same dimension, because the lack of concentration allows positive and negative contributions to partially cancel in the Euler sum while leaving large total dimensions.

8. Proof of the main theorem

We consolidate the results into a single statement.

Theorem 8.1 (Three mechanisms of concentration)**.** *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra of dimension d over a field of characteristic zero. The following are equivalent:*

(i) $\mathfrak{g}$ is semisimple.
(ii) The bracket surjectivity index $\sigma_w(\mathfrak{g}) = 1$ for all $w \geq 2$.
(iii) The Killing form $B(x,y) = \mathrm{tr}(\mathrm{ad}_x \circ \mathrm{ad}_y)$ is nondegenerate.
(iv) The CE cohomology $H^\bullet(\mathfrak{g}_-, \Bbbk)$ is concentrated in the sense that for each CE degree p, the number of weights w with $H^p(\mathfrak{g}_-, \Bbbk)_w \neq 0$ equals $|W^p_{\mathrm{aff}}|$ (the number of minimal affine Weyl coset representatives of length p).

For algebras of the same dimension d, the concentration ratio satisfies

$$\rho_w(\mathfrak{g}_{\mathrm{ab}}) \geq \rho_w(\mathfrak{g}_{\mathrm{nil}}) \geq \rho_w(\mathfrak{g}_{\mathrm{ss}})$$

at every weight w, where $\mathfrak{g}_{\mathrm{ab}}$ is abelian, $\mathfrak{g}_{\mathrm{nil}}$ is nilpotent with nontrivial bracket, and $\mathfrak{g}_{\mathrm{ss}}$ is semisimple.

Proof. The equivalence (i)⇔(ii) follows from the fact that $\sigma_w = \pi(\mathfrak{g})$ for all $w \geq 2$ (as observed in §2.1), and $\pi(\mathfrak{g}) = 1$ if and only if $\mathfrak{g}$ is perfect. For finite-dimensional Lie algebras in characteristic zero, perfectness is equivalent to semisimplicity (since the radical of a non-semisimple algebra is a proper ideal mapping to zero under the bracket projection).

The equivalence (i)⇔(iii) is Cartan's criterion.

The implication (i)⇒(iv) is the Garland–Lepowsky theorem (Theorem 1.1). For (iv)⇒(i): if $\mathfrak{g}$ is not semisimple, then the Killing form is degenerate, and Propositions 3.1 and 3.2 show that non-semisimple algebras fail to achieve the Weyl-orbit concentration pattern. Specifically, the Borel test (§3.3) demonstrates that even solvable algebras with nontrivial brackets have cohomology at weights outside the Weyl orbit loci.

The concentration ordering follows from Theorem 4.2. □

Remark 8.2. The three mechanisms are independently necessary in the following sense:

(a) The Heisenberg algebra has a nontrivial bracket ($\sigma = 1/3$) but degenerate Killing form and no Weyl group. Mechanism (I) alone is insufficient.
(b) The Borel subalgebra has $\sigma = 1/2$ (stronger than Heisenberg) but degenerate Killing form and no Weyl group. Mechanism (I) at partial strength is still insufficient.
(c) No non-semisimple algebra possesses a nondegenerate Killing form (Cartan's criterion), so mechanisms (II) and (III) are available only for semisimple algebras.

All three mechanisms engage simultaneously if and only if $\mathfrak{g}$ is semisimple.

9. Computational verification

All computations in this paper are independently verified by the `garland_lepowsky_why_engine` module (77 tests) and the `bar_loop_group_engine` module (62 tests) in the repository `compute/`. The verification follows the multi-path protocol: every numerical value is confirmed by at least two independent computation methods.

(1) The Garland–Lepowsky formula (2) is verified by direct CE matrix computation at weights $w \leq 10$ and independently by the Euler product cross-check (signed sums match $\prod(1-t^n)^3$).
(2) The concentration ratios in Table 1 are computed from CE cohomology dimensions (path 1) and independently from the ratio of total cohomology to total cochain dimensions (path 2).
(3) The Killing form determinants are computed by explicit matrix construction (ad-matrices and trace products) and independently verified by the rank computation.
(4) The Borel test (Proposition 3.2) is verified by comparing Borel and abelian total dimensions weight by weight, with the Euler series cross-check confirming that signed sums agree.
(5) The bracket surjectivity indices are computed from the rank of the bracket image matrix and independently verified to equal the perfectness ratio.
(6) The $\mathfrak{sl}_3$ bidegree data is verified by direct 8×8 CE matrix computation and independently by the Euler series cross-check ($\prod(1-t^n)^8$ coefficients).

10. Open questions

The analysis in this paper suggests several directions for further investigation.

(1) **Quantitative concentration for higher-rank algebras.** The $\mathfrak{sl}_2$ case has a single Weyl orbit per CE degree ($|W^p_{\text{aff}}| = 1$), producing the cleanest concentration. For $\mathfrak{sl}_N$ with $N \geq 3$, the affine Weyl group $W_{\text{aff}}(A_{N-1})$ has $|W^p_{\text{aff}}| > 1$ at higher lengths, so multiple weights contribute at each CE degree. The computed data for $\mathfrak{sl}_3$ shows concentration that is still strong (zero total cohomology at $w = 3$) but less sharp than $\mathfrak{sl}_2$. A precise formula for $|W^p_{\text{aff}}(A_{N-1})|$ as a function of p and N would quantify how concentration degrades with rank. For exceptional types, the orbit structure is more complex, and the concentration pattern has not been computed beyond the first few weights.
(2) **Concentration at critical level.** The Garland–Lepowsky theorem applies to the loop algebra $\mathfrak{g}_-$ independently of the level k of the affine vertex algebra $V_k(\mathfrak{g})$. At the critical level $k = -h^\vee$, the vertex algebra develops a large center (the Feigin–Frenkel center $\mathfrak{z}(\hat{\mathfrak{g}})$), and the PBW spectral sequence may have different degeneration behavior. The CE concentration persists (it depends only on $\mathfrak{g}$, not k), but the passage from CE to bar cohomology may involve higher differentials. Whether bar concentration holds at critical level is an open question with implications for the geometric Langlands programme.
(3) **Refinement of the concentration ordering.** Theorem 4.2 establishes $\rho(\text{abelian}) \geq \rho(\text{nilpotent}) \geq \rho(\text{semisimple})$. Can this be refined to a continuous ordering indexed by the perfectness ratio π? For algebras of the same dimension, is $\rho_w(\mathfrak{g})$ a monotone decreasing function of $\pi(\mathfrak{g})$ at each weight w? The computed examples support this, but the Borel and Heisenberg algebras have different dimensions (2 and 3), so a direct comparison requires normalization.

(4) **Characteristic p phenomena.** The entire analysis assumes characteristic zero. In positive characteristic, Weyl's complete reducibility theorem fails, the Killing form may be degenerate for simple algebras (the first Whitehead lemma fails), and the Garland–Lepowsky concentration may break down. The $\mathfrak{sl}_2$ concentration at weight $p(p+1)/2$ with dimension $2p+1$ suggests a connection to the representation theory of SL_2 in characteristic p (where V_p becomes reducible). An investigation of CE cohomology concentration in positive characteristic would illuminate the role of complete reducibility in the mechanism.

(5) **Infinite-dimensional generalizations.** The Virasoro algebra is perfect ($[L_m, L_n] = (m-n)L_{m+n} + \dots$ is surjective onto $\mathrm{Vir}/\mathbb{C}c$) and has a nondegenerate Shapovalov form (the analogue of the Killing form). Yet it is not semisimple in the classical sense. Whether the Virasoro loop algebra exhibits Garland–Lepowsky-type concentration is an interesting question; the shadow depth $r_{\max} = \infty$ for the Virasoro vertex algebra suggests that the concentration pattern is qualitatively different.

References

[1] H. Garland and J. Lepowsky, *Lie algebra homology and the Macdonald–Kac formulas*, Invent. Math. **34** (1976), 37–76.

[2] B. Kostant, *Lie algebra cohomology and the generalized Borel–Weil theorem*, Ann. of Math. (2) **74** (1961), 329–387.

[3] S. Kumar, *Kac–Moody Groups, their Flag Varieties and Representation Theory*, Progress in Mathematics, vol. 204, Birkhäuser, 2002.

[4] R. Bott and H. Samelson, *Applications of the theory of Morse to symmetric spaces*, Amer. J. Math. **80** (1958), 964–1029.

[5] R. Lorgat, *Modular Koszul Duality*, in preparation, 2026.

[6] E. Frenkel and D. Ben-Zvi, *Vertex Algebras and Algebraic Curves*, 2nd ed., Mathematical Surveys and Monographs, vol. 88, AMS, 2004.

[7] H. Weyl, *Theorie der Darstellung kontinuierlicher halbeinfacher Gruppen durch lineare Transformationen*, Math. Z. **23** (1925), 271–309; **24** (1926), 328–376, 377–395.

[8] V. G. Kac, *Infinite-dimensional Lie Algebras*, 3rd ed., Cambridge University Press, 1990.

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